VERITY SCOPE™

Production Product Schema (v1.2.4 (CRAFT-aligned))

Life Sciences Primary Market Research (PMR) Bid and Scope Automation

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1. Product overview

VERITY SCOPE™ is a bid and scope orchestration product for Life Sciences primary market research (PMR) teams. It turns an unstructured client brief (typically received by email) into a submit-ready bid package: scope, timeline, work breakdown structure (WBS), effort estimate, pricing pack, assumptions/risks, and proposal/SoW drafts. The product is designed for multi-company deployment, where each company has its own rate cards and pricing policies. Every extracted requirement, assumption, and sample decision is evidence-linked to the originating RFP section or attachment for defensibility and auditability.

What VERITY SCOPE™ standardizes

- A consistent study library and scoping playbooks (what a 'U&A' or 'Tracker' means in practice).
- A granular WBS task library with unit-based estimation (how effort is calculated).
- A rules engine for gap detection, clarifications, complexity multipliers, and approval routing.
- A repeatable workflow with hard gates for approvals and submission.
- Versioning and audit trail across scope, pricing, and documents.

What each company configures (tenant-specific)

- Rate cards: roles, cost rates, sell rates, geography variants, effective dates.
- Pricing policies: margin thresholds, discount limits, approval thresholds.
- Preferred vendors and OOP (out-of-pocket) catalogs: panels, recruiters, translation, incentives, transcription.
- Templates: proposal, SoW, pricing annex, assumptions annex, timeline annex.
- Email and security settings: mailboxes, allowed domains, retention, access controls.

Primary outcomes

- Faster bid turnaround with fewer rework cycles.
- Higher consistency and traceability of effort and pricing.
- Clear assumptions and risks to reduce scope creep and delivery surprises.
- Comparable operational metrics across bids and teams (cycle time, win rate, margin accuracy).

2. Key product concepts

Core objects

- Opportunity: A single bid instance (client request) with a due date and owner.
- Brief: Structured extraction of the client request (objectives, markets, audience, sample, deliverables).
- Study definition: A standardized description of a study type (required inputs, default deliverables, WBS mapping).
- Scope: The finalized scope statement (method, sample plan, sample size recommendation, HCP shortlist, deliverables, assumptions).
- Workplan: Timeline and milestones, including explicit client review windows.
- WBS: Task-level work breakdown structure used for effort and cost calculations.
- Pricing pack: Labor + OOP + policies + margin checks; optionally tiered options (Good/Better/Best).
- Documents: Proposal/SoW/annexures generated from templates and version-controlled.
- Approvals: Configurable routing and sign-off requirements (finance, legal, compliance).
- Submission: The outbound package (usually an email draft) and its audit record.

Standardization vs configuration

VERITY SCOPE™ uses a global effort model (standard tasks, units, base hours, multipliers) and applies company-specific pricing via configurable rate cards and policies. This keeps estimation logic consistent while allowing each company to reflect its own economics.

Email-first design

- Inbound: email thread and attachments are the default intake channel.
- Outbound: the product generates a submission-ready email draft with the correct attachments; a human user sends it.
- Optional: structured web intake form, portal upload, CRM sync, procurement system sync (e.g., Coupa).

3. Personas, roles, and permissions

VERITY SCOPE™ supports role-based access control (RBAC). Permissions are assigned per tenant.

Primary personas

Persona	Primary responsibility	Typical actions in Verity SCOPE
Bid Lead / Bid Desk	Owns bid response and submission	Review brief, approve assumptions, approve scope/pricing, submit bid
Research Lead	Owns research design quality	Confirm methodology, sample plan, analysis approach, deliverables
Project Manager (PM)	Owns timeline and execution plan	Build workplan, manage review cycles, ensure feasibility
Analyst / Associate	Supports content and analysis	Draft sections, build tab plan, charting, QC
Programmer / Scripting	Builds survey scripts (quant)	Confirm LOI/logic complexity assumptions, validate programming effort
Fieldwork / Operations	Recruitment and field execution	Feasibility checks, vendor coordination, QC pacing
Finance approver	Protects margin and discount policy	Approve exceptions, approve discounts, margin overrides
Legal / Compliance	Contract, privacy, and safety controls	Approve SoW deviations, consent language, data handling requirements
Tenant Admin	Configuration and governance	Manage rate cards, templates, policies, integrations, users

Permission groups (example)

- Viewer: read-only access to opportunities and artifacts.
- Contributor: can edit brief, scope, and WBS but cannot approve or submit.
- Approver: can approve specific gates (scope, pricing, legal, compliance).
- Bid Owner: can move opportunity through workflow, generate documents, and prepare submission.
- Admin: tenant configuration, rate cards, templates, policies, integrations, user management.

4. Product modules

Modules are designed so teams can start email-first and add integrations later without reworking the core estimation logic.

Module	Purpose	Key outputs
Intake Hub	Connects to email/portal/CRM, ingests briefs and attachments, extracts structured brief fields, and tracks the email thread.	Opportunity, raw intake bundle, structured Brief
Study Library	Standardized PMR study taxonomy, required inputs, playbooks, default deliverables, and task mappings.	Study definitions, playbooks, task sets
Clarifications Engine	Completeness scoring, gap detection, question selection, assumptions and risks register.	Completeness score, clarifications email draft, assumptions/risks
Scope Builder	Methodology, sample plan, quotas, sample size recommendation, HCP shortlist, deliverables, and scope narrative builder.	Scope statement, sample plan, sample size options, HCP shortlist (primary + backup), deliverables list
Workplan Builder	Milestones, timelines, dependencies, and explicit client review windows.	Workplan milestones, timeline
Estimation Engine	WBS task selection, units calculation, base hours, multipliers, and audit trail.	WBS lines, effort summary, multiplier log
Pricing Engine	Tenant rate cards, OOP catalogs, margin checks, discount policies, tiered options.	Pricing pack, margin checks, tier options
Docs & Templates	Proposal/SoW/annex generation, versioning, and consistency QC checks.	Proposal, SoW, annexures, QC report
Approvals & Governance	Routing, sign-off, exception handling, and approval logs.	Approval requests, decisions, exception logs
Submission & Logging	Email draft creation (default) or send-via-mailbox (optional), submission logs.	Submission email draft, submission record
Win Handoff	Locks scope and exports delivery plan and staffing demand into delivery tools.	Delivery handoff pack, project plan export
Admin & Configuration	Tenant setup: rate cards, policies, vendors, templates, integrations, user permissions.	Tenant configuration, policy versioning
Analytics & Reporting	Cycle time, win-rate, margin accuracy, rework rates, throughput metrics.	Dashboards, exports

5. Multi-tenant configuration model

VERITY SCOPE™ is multi-tenant: each company (tenant) has isolated data and configuration. Global libraries provide defaults, and each tenant can override where needed.

Configuration layers

- Global (product-managed): study taxonomy, baseline task library, baseline multipliers, baseline question library.
- Tenant default: rate cards, templates, pricing policies, vendor catalogs, approval routing rules.
- Client-specific (optional): client-specific assumptions, preferred deliverables, client template variants.
- Opportunity-specific: bid-level assumptions, rush flags, custom scope clauses.

Override precedence (highest wins)

- Opportunity override
- Client override (optional)
- Tenant default
- Global default

Tenant configuration checklist

- Rate cards: roles, cost and sell rates, geo bands, effective dates.
- Pricing policies: margin thresholds, discount rules, approval triggers.
- OOP catalogs: panels, recruiters, incentives rules, translations, transcription, software.
- Document templates: proposal/SoW/pricing annex; branding and clause sets.
- Email settings: connected inbox, allowed sender domains, signature blocks, attachment limits.
- User roles and permissions: RBAC groups and approval authority.
- Integrations: CRM/procurement/storage connectors and field mappings.

6. End-to-end workflow (state machine)

The product enforces a controlled workflow. Each status has required fields and allowed transitions. This prevents incomplete bids and makes approvals auditable.

Status	Meaning	Required to enter	Key outputs
NEW_INTAKE	Opportunity created; raw email/files captured.	opp_id, source, owner, due_date	Raw intake bundle stored
BRIEF_PARSED	Brief fields extracted into structured format.	brief fields (partial allowed) + confidence scores	Structured Brief v1
BRIEF_GAPS_FOUND	Critical inputs missing; clarifications required.	missing fields list	Clarification list + assumptions draft
CLARIFICATIONS_SENT	Clarification email sent (human action).	clarification questions logged	Email draft + sent timestamp
SCOPE_DRAFTED	Scope, deliverables, and workplan drafted.	method, sample plan, deliverables, workplan	Scope v1 + Workplan v1
WBS_GENERATED	Task-level WBS created and unitized.	scope + drivers	WBS v1 + multiplier log
COSTED	Labor and OOP costed; margin checked.	WBS + rate card + OOP items	Pricing pack v1 + exceptions
DOCS_DRAFTED	Proposal/SoW drafts generated and QC checked.	scope + pricing + templates	Docs v1 + QC report
INTERNAL_REVIEW	Approvals in progress (scope/pricing/legal).	approval requests created	Approval decisions + comments
REWORK_REQUIRED	Rework requested; revisions needed before approval.	Rejection/hold decision + reason logged	Rework tasks + version increment
APPROVED_TO_SUBMIT	All required approvals complete.	all required approvals = approved	Final bid pack locked
SUBMITTED	Bid submitted (usually by email).	submission record	Submitted timestamp + recipients
WON_HANDOFF	Bid won; delivery handoff created.	won flag + final scope	Handoff pack + project setup
LOST_ARCHIVE	Bid lost; archived with reason.	lost reason (optional)	Archive record
CANCELLED	Bid cancelled before submission.	cancel reason	Archive record

Allowed transitions (simplified)

NEW_INTAKE -> BRIEF_PARSED -> (BRIEF_GAPS_FOUND -> CLARIFICATIONS_SENT -> BRIEF_PARSED) -> SCOPE_DRAFTED -> WBS_GENERATED -> COSTED -> DOCS_DRAFTED -> INTERNAL_REVIEW -> (APPROVED_TO_SUBMIT or REWORK_REQUIRED) -> SUBMITTED -> (WON_HANDOFF or LOST_ARCHIVE)

If INTERNAL_REVIEW results in REWORK_REQUIRED, the bid returns to COSTED (pricing changes) or DOCS_DRAFTED (document edits) based on rejection reason. Each loop increments version and writes an AuditEvent.

7. Data model (production schema)

This section defines the production-grade schema at the level needed for engineering implementation. Field names are written in snake_case. Types are logical (implementation may vary by database). All entities include created/updated timestamps and a version field where relevant.

7.1 Tenant and access control

Tenant

Represents a company using VERITY SCOPE™. All data is isolated per tenant.

Field	Type	Req.	Notes
tenant_id	string (UUID)	Yes	Primary key.
name	string	Yes	Company name.
status	enum	Yes	active | suspended | trial.
default_timezone	string	Yes	IANA timezone, e.g., Asia/Kolkata.
data_retention_days	int	No	Optional policy.
created_at	datetime	Yes	System-generated.
updated_at	datetime	Yes	System-generated.

User

A named user under a tenant.

Field	Type	Req.	Notes
user_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	Yes	FK to Tenant.
email	string	Yes	Unique within tenant.
display_name	string	Yes	Full name.
status	enum	Yes	active | invited | disabled.
role_ids	string[]	Yes	Assigned RBAC roles.
created_at	datetime	Yes	
updated_at	datetime	Yes	

Role

RBAC role composed of permissions.

Field	Type	Req.	Notes
role_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	Yes	Tenant-scoped roles; global roles may be supported.
name	string	Yes	e.g., Bid Owner, Approver, Admin.
permissions	string[]	Yes	Permission keys.
created_at	datetime	Yes	
updated_at	datetime	Yes	

ClientAccount

Optional: represents end-clients within a tenant for client-level overrides and reporting.

Field	Type	Req.	Notes
client_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	Yes	FK to Tenant.
client_name	string	Yes	
industry_segment	string	No	Optional segmentation field.
preferred_template_set_id	string	No	Client-level template override.
notes	string	No	Non-sensitive internal notes.
created_at	datetime	Yes	
updated_at	datetime	Yes	

7.2 Opportunity and communications

Opportunity

A single bid instance created from an inbound brief. It is the root object for the bid workflow.

Field	Type	Req.	Notes
opp_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	Yes	FK to Tenant.
client_id	string (UUID)	No	FK to ClientAccount (optional).
title	string	Yes	Short description of the bid.
owner_user_id	string (UUID)	Yes	Bid owner.
status	enum	Yes	Workflow state (see Section 6).
due_date	date	Yes	Bid submission due date.
study_family	enum	No	Assigned after classification.
study_type	enum	No	Assigned after classification.
method	enum	No	quant | qual | mixed.
priority	enum	No	P0 | P1 | P2.
source	enum	Yes	email | portal | form | crm | api.
metadata	map<string,string>	No	Optional intake/procurement fields (e.g., cost_center, project_code) used for integrations and reporting.
version	int	Yes	Increment on major revisions.
created_at	datetime	Yes	
updated_at	datetime	Yes	

EmailThread

Stores inbound and outbound email context for auditability and threading.

Field	Type	Req.	Notes
thread_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	FK to Opportunity.
channel	enum	Yes	gmail | outlook | smtp_forward.
external_thread_id	string	No	Provider thread ID.
subject	string	Yes	
participants	string[]	Yes	Email addresses.

last_message_at	datetime	Yes	
created_at	datetime	Yes	
updated_at	datetime	Yes	

EmailMessage

A single email in a thread. Body may be stored as sanitized HTML + plaintext.

Field	Type	Req.	Notes
message_id	string (UUID)	Yes	Primary key.
thread_id	string (UUID)	Yes	FK to EmailThread.
direction	enum	Yes	inbound | outbound_draft | outbound_sent.
from	string	Yes	Email address.
to	string[]	Yes	
cc	string[]	No	
sent_at	datetime	No	Set when outbound is sent.
body_text	string	No	Plaintext.
body_html	string	No	Sanitized HTML.
created_at	datetime	Yes	
updated_at	datetime	Yes	

Attachment

Files attached to an email or uploaded through other intake channels.

Field	Type	Req.	Notes
attachment_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
source_message_id	string (UUID)	No	FK to EmailMessage when applicable.
file_name	string	Yes	
mime_type	string	Yes	application/pdf, application/vnd.openxmlformats-officedocument.wordprocessingml.document, etc.
file_size_bytes	int	Yes	
storage_uri	string	Yes	Object storage pointer.
checksum_sha256	string	No	Optional integrity check.
created_at	datetime	Yes	
updated_at	datetime	Yes	

7.3 Brief, clarifications, assumptions, and risks

Brief

Structured representation of the client request. Populated via parsing and human edits.

Field	Type	Req.	Notes
brief_id	string (UUID)	Yes	Primary key.
opp_id	string (UUID)	Yes	FK to Opportunity.
objectives	string[]	No	Top objectives in client language.
background	string	No	Relevant context.
audiences	object[]	No	Audience definitions (see Audience object below).
countries	string[]	No	ISO country codes or names.
languages	string[]	No	Language codes; can be derived from countries.
method_preference	enum	No	quant | qual | mixed | unspecified.
sample_total_target	int	No	Quant total completes target.
sample_by_country	map<string,int>	No	Country -> completes.
quotas	object	No	Quota structure; optional for many studies.
loi_minutes	int	No	Quant length of interview.
session_length_minutes	int	No	Qual session length.
stimuli	object	No	Stimuli count, format, readiness.
advanced_analytics	string[]	No	Flags: conjoint, maxdiff, segmentation, drivers, etc.
deliverables_requested	string[]	No	Client-stated deliverables (may differ from standard).
fieldwork_window_start	date	No	
fieldwork_window_end	date	No	
compliance_constraints	string[]	No	GDPR, consent, adverse event process, etc.
budget_range	string	No	Optional, may be redacted.
completeness_score	int	Yes	0-100.
confidence_by_field	map<string,enum>	No	Field -> high | medium | low.
created_at	datetime	Yes	
updated_at	datetime	Yes	

Audience (embedded object)

Field	Type	Req.	Notes
audience_type	enum	Yes	hcp | patient | caregiver | payer | pharmacist | nurse | admin | other.
name	string	Yes	e.g., Cardiologists, Type 2 diabetes patients.
inclusion_criteria	string[]	No	
exclusion_criteria	string[]	No	
quota_rules	object	No	Optional per-audience quota rules.
recruit_difficulty	enum	No	easy | medium | hard | rare.

ClarificationQuestionDefinition

A library question used for gap detection and clarifications emails.

Field	Type	Req.	Notes
question_id	string	Yes	Stable ID, e.g., UAA-QUANT-LOI-01.
study_type	enum	Yes	Study type this question belongs to.
method	enum	No	quant | qual | mixed; optional.
section	string	Yes	e.g., Sample, Instrument, Deliverables.
critical	bool	Yes	Critical questions block submission unless answered or assumed.
question_text	string	Yes	
mapped_field	string	Yes	Target field path in Brief/Scope.
trigger_when	string	No	Rule expression to auto-add question.
default_assumption	string	No	Used when client cannot respond in time.
active	bool	Yes	Enable/disable in library.

ClarificationInstance

A question instance linked to an opportunity.

Field	Type	Req.	Notes
clarification_id	string (UUID)	Yes	Primary key.
opp_id	string (UUID)	Yes	
question_id	string	Yes	FK to ClarificationQuestionDefinition.
status	enum	Yes	open | answered | assumed | waived.

answer_text	string	No	Client response or internal decision.
resolved_at	datetime	No	
resolved_by_user_id	string (UUID)	No	

AssumptionRisk

Tracks assumptions and associated risks. Assumptions must be included in the submission package.

Field	Type	Req.	Notes
risk_id	string (UUID)	Yes	Primary key.
opp_id	string (UUID)	Yes	
assumption_text	string	Yes	
impact_area	enum	Yes	scope | timeline | cost | compliance | quality.
risk_level	enum	Yes	low | medium | high.
mitigation	string	No	
status	enum	Yes	open | accepted | mitigated.
created_by_user_id	string (UUID)	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

7.4 Study library and playbooks

StudyFamily

Top-level MECE families. Each opportunity must have exactly one primary family.

Field	Type	Req.	Notes
study_family	enum	Yes	Canonical keys stored in DB/API. Display labels shown in UI. UNDERSTANDING_DIAGNOSIS (Understanding & Diagnosis) | TRACKING_MONITORING (Tracking & Monitoring) | TESTING_OPTIMIZATION (Testing & Optimization) | TRADEOFF_CHOICE (Trade-off & Choice) | SEGMENTATION_TARGETING (Segmentation & Targeting) | PRICING_MARKET_ACCESS (Pricing & Market Access).
description	string	Yes	
active	bool	Yes	

StudyType

A named study type under a family (e.g., U&A, Tracker, Concept Test).

Field	Type	Req.	Notes
study_type	enum	Yes	Stable enum key.
study_family	enum	Yes	FK to StudyFamily.
name	string	Yes	Display name.
default_method	enum	Yes	quant | qual | mixed.
supported_modes	string[]	Yes	CAWI, CATI, IDI, FGD, community, diary, etc.
active	bool	Yes	

StudyDefinition

The production definition of a study type: required fields, drivers, default deliverables, and mappings.

Field	Type	Req.	Notes
study_definition_id	string (UUID)	Yes	Primary key.
study_type	enum	Yes	
version	int	Yes	Allows library evolution while keeping historical bids reproducible.
required_fields	string[]	Yes	List of required field paths.
driver_fields	string[]	Yes	Fields that impact effort (countries, languages, LOI, etc.).

default_deliverables	string[]	Yes	References DeliverableDefinition IDs.
default_task_set_id	string (UUID)	Yes	References TaskSet.
default_multiplier_set_id	string (UUID)	Yes	References MultiplierSet.
default_question_set_id	string (UUID)	Yes	References QuestionSet.
default_workplan_template_id	string (UUID)	Yes	References WorkplanTemplate.
active	bool	Yes	

DeliverableDefinition

Standard deliverables offered across PMR studies.

Field	Type	Req.	Notes
deliverable_id	string	Yes	Stable ID, e.g., TABBOOK, TOPLINE_DECK.
name	string	Yes	Display name.
description	string	Yes	Plain language definition.
acceptance_criteria	string[]	No	What is considered 'done'.
default_inclusions	string[]	No	Optional: standard inclusions.
active	bool	Yes	

TemplateSet

A bundle of document templates used for generation (tenant or client-specific).

Field	Type	Req.	Notes
template_set_id	string (UUID)	Yes	
tenant_id	string (UUID)	No	Null means global default.
name	string	Yes	
proposal_template_id	string (UUID)	No	
sow_template_id	string (UUID)	No	
pricing_annex_template_id	string (UUID)	No	
assumptions_annex_template_id	string (UUID)	No	
timeline_annex_template_id	string (UUID)	No	
active	bool	Yes	

7.5 Scope, sample plan, and workplan

Scope

The scoped solution to deliver. Generated from playbooks and edited by users.

Field	Type	Req.	Notes
scope_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
method	enum	Yes	quant | qual | mixed.
study_type	enum	Yes	
scope_summary	string	No	Plain language summary.
sample_plan	object	Yes	See SamplePlan.
deliverables	string[]	Yes	Deliverable IDs.
assumptions	string[]	No	Human-readable assumptions included in docs.
exclusions	string[]	No	Out-of-scope items to avoid ambiguity.
compliance_notes	string[]	No	
version	int	Yes	
created_at	datetime	Yes	
updated_at	datetime	Yes	

SamplePlan

A structured plan for who will be interviewed and where.

Field	Type	Req.	Notes
audiences	object[]	Yes	Audience objects.
countries	string[]	Yes	
cells	object[]	Yes	List of SampleCell objects.
total_n	int	No	Derived sum for quant.
notes	string	No	Feasibility notes.

SampleCell (embedded object)

Field	Type	Req.	Notes
country	string	Yes	
audience_name	string	Yes	References an Audience in SamplePlan.audiences.
target_n	int	Yes	Completes (quant) or sessions (qual).
mode	string	No	e.g., CAWI, CATI, IDI, FGD.
quota_rules	object	No	Optional.
recruit_source	enum	No	panel | list | recruiter | sites | mixed.
recruit_difficulty	enum	No	easy | medium | hard |

			rare.
incentive_amount	number	No	Optional; currency in tenant config.

SampleSizeRecommendation

Recommended sample size options for the opportunity, with rationale, feasibility impacts, and evidence links. Supports minimum/recommended/high-confidence bid options.

Field	Type	Req.	Notes
recommendation_id	string (UUID)	Yes	Primary key.
opp_id	string (UUID)	Yes	
options	object[]	Yes	List of sample size options (minimum | recommended | high_confidence). Each option includes option_code, total_n, cells/quota impacts, timeline_impact_days, cost_impact_percent, assumptions_used.
selected_option	string	Yes	Selected option code (or option_id).
confidence_score	number	No	0-1 confidence score.
rationale	string	No	Plain-language rationale for the recommendation.
evidence	object[]	No	FieldEvidence refs for RFP-stated constraints (e.g., required N, segment quotas, timeline).
created_at	datetime	Yes	
updated_at	datetime	Yes	

HCPUniverseQuery

A stored query receipt for how the eligible HCP universe was filtered and counted (aggregated only). Used to justify feasibility and shortlist sizing.

Field	Type	Req.	Notes
query_id	string (UUID)	Yes	Primary key.
opp_id	string (UUID)	Yes	
audience_name	string	Yes	Matches SamplePlan.audiences[].name.
countries	string[]	Yes	
filters	object	Yes	Eligibility filters (specialty, setting, patient volume, institution type, language, etc.).

exclusions	object	No	Exclusions/blocks (blacklists, conflicts, compliance).
result_counts	object	Yes	Eligible counts by country/segment (aggregated).
evidence	object[]	No	FieldEvidence refs for RFP-stated targeting constraints.
created_at	datetime	Yes	
updated_at	datetime	Yes	

HCPShortlist

A ranked shortlist of eligible HCPs selected from an internal database for recruitment planning. Includes a primary and backup pool to protect feasibility.

Field	Type	Req.	Notes
shortlist_id	string (UUID)	Yes	Primary key.
opp_id	string (UUID)	Yes	
audience_name	string	Yes	Matches SamplePlan.audiences[].name.
countries	string[]	Yes	
target_n	int	Yes	Planned completes/sessions for this shortlist.
primary_hcp_ids	string[]	Yes	Internal HCP IDs only (no direct PII).
backup_hcp_ids	string[]	No	Backup pool to protect feasibility.
scoring_model_version	string	No	Version/label of the ranking logic used.
selection_rationale	string	No	Plain-language explanation of selection criteria and quota balancing.
status	enum	Yes	draft | approved.
approved_by_user_id	string (UUID)	No	
approved_at	datetime	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

Workplan

Timeline and milestone plan for execution. Used for both bid and delivery handoff.

Field	**Type**	**Req.**	**Notes**
workplan_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
milestones	object[]	Yes	List of Milestone objects.
fieldwork_days	int	No	Derived or specified.
client_review_days	int	No	Explicitly accounted for.
risk_buffer_days	int	No	Optional.
version	int	Yes	

created_at	datetime	Yes	
updated_at	datetime	Yes	

Milestone (embedded object)

Field	Type	Req.	Notes
milestone_id	string	Yes	Stable within workplan version.
name	string	Yes	e.g., Kickoff, Soft launch, Readout.
start_date	date	No	May be relative before dates are locked.
end_date	date	No	
duration_days	int	Yes	
dependencies	string[]	No	Milestone IDs.
owner_role	string	No	PM, Research Lead, etc.
notes	string	No	

7.6 Estimation and WBS

TaskDefinition

Atomic work item used to build a WBS. Tasks are selected by rules and multiplied by units and complexity.

Field	Type	Req.	Notes
task_code	string	Yes	Stable code, e.g., QNT-PROG-PROJECT_SETUP.
name	string	Yes	
phase	enum	Yes	bid | instrument | programming | fieldwork | data | analysis | reporting | pm | governance.
applies_when	string	No	Rule expression evaluated on opportunity context.
unit_type	enum	Yes	per_opp | per_country | per_language | per_country_language | per_session | per_field_day | per_cycle | per_100_verbatims | etc.
units_expr	string	Yes	Expression returning a number, e.g., countries * languages.
base_hours_by_role	map<string,number>	Yes	Role -> base hours per unit.
multiplier_codes	string[]	No	Multiplier codes to apply.
dependencies	string[]	No	Task codes.
outputs	string[]	No	Artifact names for traceability.
active	bool	Yes	

TaskSet

A named set of tasks used by a study definition (baseline WBS).

Field	Type	Req.	Notes
task_set_id	string (UUID)	Yes	
name	string	Yes	
study_type	enum	No	Optional scope.
task_codes	string[]	Yes	List of TaskDefinition.task_code.
version	int	Yes	
active	bool	Yes	

WBSLine

Computed line item used for effort and pricing. Derived from TaskDefinition + opportunity context.

Field	Type	Req.	Notes
wbs_line_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
task_code	string	Yes	
role	string	Yes	Standard role name.
unit_type	string	Yes	
units	number	Yes	
base_hours	number	Yes	Hours per unit before multipliers.
multipliers_applied	object[]	No	List of MultiplierApplied refs.
total_hours	number	Yes	
rate_ref	string	No	Pointer to tenant rate used.
labor_cost	number	No	Cost rate * hours.
labor_price	number	No	Sell rate * hours.
notes	string	No	

MultiplierDefinition

Defines a complexity factor and where it applies.

Field	Type	Req.	Notes
multiplier_code	string	Yes	e.g., M_COUNTRIES, M_LOI, M_RUSH.
driver	string	Yes	Name of driver, e.g., countries_count.
levels	object[]	Yes	List of {when, value}.
apply_to_task_prefixes	string[]	Yes	Task code prefixes this multiplier applies to.
active	bool	Yes	

MultiplierApplied

Computed record showing which multipliers were applied for an opportunity and why.

Field	Type	Req.	Notes
multiplier_applied_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
multiplier_code	string	Yes	
value	number	Yes	
reason	string	Yes	Human-readable explanation.
created_at	datetime	Yes	

7.7 Pricing (tenant-configurable)

RateCard

Tenant-specific rate card. Supports multiple versions and effective dates.

Field	Type	Req.	Notes
rate_card_id	string (UUID)	Yes	
tenant_id	string (UUID)	Yes	
name	string	Yes	e.g., 2026 Standard Rates - India.
currency	string	Yes	ISO currency code.
effective_from	date	Yes	
effective_to	date	No	
rate_items	object[]	Yes	List of RateItem objects.
active	bool	Yes	

RateItem (embedded object)

Field	Type	Req.	Notes
standard_role	string	Yes	e.g., Research Lead.
geo_band	string	No	Optional geo segmentation.
cost_rate_per_hour	number	Yes	
sell_rate_per_hour	number	Yes	
notes	string	No	

OOPCatalogItem

Catalog of out-of-pocket costs used in bids. Tenant-specific.

Field	Type	Req.	Notes
oop_item_id	string (UUID)	Yes	
tenant_id	string (UUID)	Yes	
category	enum	Yes	panel | recruiter | incentive | translation | transcription | facility | software | other.
name	string	Yes	
unit_type	enum	Yes	per_complete | per_session | per_language | per_hour | flat_fee | etc.
default_unit_cost	number	Yes	
vendor_id	string (UUID)	No	Optional preferred vendor.
notes	string	No	
active	bool	Yes	

Vendor

Preferred vendor list (optional).

Field	Type	Req.	Notes
vendor_id	string (UUID)	Yes	
tenant_id	string (UUID)	Yes	
name	string	Yes	
services	string[]	Yes	panel, translation, recruiter, etc.
regions	string[]	No	
contact_email	string	No	
active	bool	Yes	

PricingLine

A computed line in the pricing pack. Includes labor and OOP lines.

Field	Type	Req.	Notes
pricing_line_id	string (UUID)	Yes	
pricing_pack_id	string (UUID)	No	Optional FK to PricingPack. Null indicates default/base pack.
line_type	enum	Yes	labor | oop | adjustment.
name	string	Yes	Display name.
quantity	number	Yes	Hours (labor) or units (OOP).
unit_price	number	Yes	Sell price per unit.
unit_cost	number	No	Cost per unit (optional).
total_price	number	Yes	
total_cost	number	No	
linked_wbs_line_ids	string[]	No	Traceability.
notes	string	No	

PricingSummary

Summary totals and policy checks.

Field	**Type**	**Req.**	**Notes**
pricing_pack_id	string (UUID)	No	Optional FK to PricingPack. Null indicates default/base pack.
currency	string	Yes	
labor_price_total	number	Yes	
oop_price_total	number	Yes	
total_price	number	Yes	
total_cost	number	No	
gross_margin_percent	number	No	
discount_percent	number	No	
exceptions	string[]	No	Policy exceptions

			requiring approvals.

7.8 Approvals and governance

ApprovalPolicy

Tenant configuration defining when approvals are required.

Field	Type	Req.	Notes
approval_policy_id	string (UUID)	Yes	
tenant_id	string (UUID)	Yes	
name	string	Yes	
rules	object[]	Yes	List of approval rules (see below).
version	int	Yes	
active	bool	Yes	

ApprovalRule (embedded object)

Field	Type	Req.	Notes
rule_id	string	Yes	
applies_when	string	Yes	Rule expression on opportunity/pricing context.
approval_type	enum	Yes	scope | pricing | discount | legal | compliance.
required_role	string	Yes	RBAC role required to approve.
sla_hours	int	No	Optional internal target.
notes	string	No	

ApprovalRequest

An approval instance created for an opportunity.

Field	Type	Req.	Notes
approval_request_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
approval_type	enum	Yes	
status	enum	Yes	pending | approved | rejected | cancelled.
requested_by_user_id	string (UUID)	Yes	
requested_at	datetime	Yes	
decisions	object[]	No	List of ApprovalDecision.
comments	string	No	

ApprovalDecision (embedded object)

Field	Type	Req.	Notes
decided_by_user_id	string (UUID)	Yes	
decision	enum	Yes	approved | rejected.
decision_at	datetime	Yes	
comment	string	No	Reason or conditions.

7.9 Documents, templates, and QC

DocumentTemplate

A template file with placeholders used for generation (tenant-specific or global).

Field	Type	Req.	Notes
template_id	string (UUID)	Yes	
tenant_id	string (UUID)	No	Null for global default.
name	string	Yes	
doc_type	enum	Yes	proposal | sow | pricing_annex | assumptions_annex | timeline_annex.
file_uri	string	Yes	Template file storage pointer.
placeholder_schema_version	int	Yes	Version of placeholder tags supported.
active	bool	Yes	
created_at	datetime	Yes	
updated_at	datetime	Yes	

DocumentArtifact

A generated document version for an opportunity.

Field	Type	Req.	Notes
artifact_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
doc_type	enum	Yes	
version	int	Yes	Document version per type.
file_uri	string	Yes	Generated file storage pointer.
generated_at	datetime	Yes	
generated_by_user_id	string (UUID)	No	
status	enum	Yes	draft | final.
qc_report_id	string (UUID)	No	Link to QC report.

QCReport

Automated consistency checks to prevent common bid errors.

Field	Type	Req.	Notes
qc_report_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
checks	object[]	Yes	List of QCCheck results.
overall_status	enum	Yes	pass | warn | fail.
created_at	datetime	Yes	

QCCheck (embedded object)

Field	Type	Req.	Notes

check_code	string	Yes	e.g., QC_SAMPLE_MATCH.
status	enum	Yes	pass | warn | fail.
message	string	Yes	Human-readable detail.
related_fields	string[]	No	Field paths involved.

7.10 Submission and win handoff

SubmissionRecord

Captures how and when a bid was submitted.

Field	Type	Req.	Notes
submission_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
channel	enum	Yes	email_draft | email_sent | portal_upload | api.
recipients	string[]	No	Email recipients when applicable.
subject	string	No	Email subject.
artifact_ids	string[]	Yes	Documents included.
submitted_at	datetime	No	Set when actually sent.
submitted_by_user_id	string (UUID)	No	
notes	string	No	

HandoffPack

Created when an opportunity is marked as won. Used to start delivery with minimal rework.

Field	Type	Req.	Notes
handoff_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
locked_scope_version	int	Yes	
locked_pricing_version	int	Yes	
workplan_version	int	Yes	
resource_demand	object	No	Role -> hours by week (optional).
export_targets	string[]	No	Delivery tools to push to.
created_at	datetime	Yes	
baseline_package_id	string (UUID)	Yes	Reference to BaselinePackage snapshot used by VERITY CRAFT (created at scope baseline lock / win handoff).

7.10.1 Baseline package (VERITY CRAFT integration)

BaselinePackage is the immutable snapshot of signed-off scope inputs that VERITY CRAFT consumes. For SCOPE-connected delivery, it is created when the delivery baseline is locked (typically at WON_HANDOFF after win) and referenced by the event scope.baseline_locked.

BaselinePackage

Field	Type	Req.	Notes
baseline_package_id	string (UUID)	Yes	Primary identifier; used as baseline_package_ref

			in integrations.
opp_id	string (UUID)	Yes	Opportunity / bid identifier.
source_scope_version	int	Yes	Pinned scope version at baseline lock.
source_pricing_version	int	Yes	Pinned pricing version at baseline lock.
source_workplan_version	int	Yes	Pinned workplan / WBS version at baseline lock.
classification	object	No	Outputs of scope.study_classified (study type, modality, complexity tags).
study_definition_version_ref	ref	Yes	Pinned StudyDefinition (id + version). Join key for StudyDefinitionBinding used by CRAFT to auto-select blueprint defaults.
study_parameters	object	Yes	Countries, languages, audiences, LOI/session length, waves, method (qual/quant/hybrid).
sample_plan	object	No	Selected sample size option + quotas (if applicable).
deliverables	object	Yes	Expected outputs (e.g., report, deck, data file, transcripts, audio/video requirements).
constraints	object	No	Timing, compliance, recruitment, stimulus constraints.
pinned_artifact_refs	object	No	Refs to pinned Scope Pack artifacts (SoW/proposal annexures) as needed for instrument build.
created_at	datetime	Yes	
created_by	string	Yes	User/service who locked baseline.

7.10.2 Scope change control (ChangeOrder)

ChangeOrder captures any post-baseline scope change request (including those raised from VERITY CRAFT when instrument constraints surface). Approval results in a new scope version and a new BaselinePackage; downstream packs are marked out_of_date until repinned.

ChangeOrder

Field	Type	Req.	Notes
change_order_id	string (UUID)	Yes	
opp_id	string (UUID)	Yes	
baseline_package_id	string (UUID)	Yes	Baseline being changed.
requested_by	string	Yes	User/service; may be VERITY CRAFT.
change_summary	string	Yes	Plain-language description of requested change.
delta	object	No	Structured delta (e.g., countries, languages, audience, LOI, quotas, deliverables).
status	enum	Yes	DRAFT | SUBMITTED | APPROVED | REJECTED | CANCELLED.
approval_type	enum	Yes	scope (and optionally pricing/legal/compliance depending on policy).
created_at	datetime	Yes	
approved_at	datetime	No	
resulting_scope_version	int	No	Populated when approved; increments scope version.
resulting_baseline_package_id	string (UUID)	No	New BaselinePackage created on approval.

7.10.3 Integration events (external, platform bus)

The following platform events are emitted for downstream systems (e.g., VERITY CRAFT). These are distinct from internal AuditEvent logs.

Implementation Decision (MVP): study_id is the canonical identifier for all SCOPE↔CRAFT platform events (opp_id, if present, is a legacy alias only).

- scope.study_classified: Emitted after brief parsing/classification; includes study_id, study_definition_version_ref, and classification.
- scope.baseline_locked: Emitted when delivery baseline is locked and BaselinePackage is created; includes study_id, baseline_package_ref, and artifact_refs[] (pinned Scope pack artifact versions).
- scope.change_order_approved: Emitted when a ChangeOrder is approved; includes study_id, resulting_scope_version, and new_baseline_package_ref.

7.11 Audit events

AuditEvent

Immutable audit trail for workflow actions, approvals, and document generation.

Field	**Type**	**Req.**	**Notes**

audit_event_id	string (UUID)	Yes	
tenant_id	string (UUID)	Yes	
opp_id	string (UUID)	No	Null for tenant-level events.
event_type	string	Yes	e.g., BRIEF_PARSED, PRICING_APPROVED.
actor_user_id	string (UUID)	No	Null for system actions.
event_at	datetime	Yes	
metadata	object	No	Non-sensitive event details.
ip_address	string	No	Optional.
user_agent	string	No	Optional.

7.15 Library sets, templates, and provenance (v1.2 additions)

MultiplierSet			
Field	Type	Req.	Notes
multiplier_set_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	No	Null means global default.
name	string	Yes	
study_type	enum	No	Optional scope.
multiplier_codes	string[]	Yes	List of MultiplierDefinition.multiplier_code.
version	int	Yes	Increment on changes.
active	bool	Yes	
QuestionSet			
Field	Type	Req.	Notes
question_set_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	No	Null means global default.
name	string	Yes	
study_type	enum	No	Optional scope.
question_ids	string[]	Yes	List of ClarificationQuestionDefinition.question_id.
version	int	Yes	Increment on changes.
active	bool	Yes	
WorkplanTemplate			
Field	Type	Req.	Notes
workplan_template_id	string (UUID)	Yes	Primary key.
tenant_id	string (UUID)	No	Null means global default.
name	string	Yes	
study_type	enum	No	Optional scope.
milestones_template	object[]	Yes	List of MilestoneTemplate objects.
version	int	Yes	Increment on changes.
active	bool	Yes	
MilestoneTemplate			
Field	Type	Req.	Notes
milestone_template_id	string	Yes	Stable within template.
name	string	Yes	e.g., Kickoff, Soft launch, Readout.
sequence	int	Yes	Sort order for UI.
duration_days	int	Yes	
depends_on	string[]	No	milestone_template_id dependencies.
owner_role	string	No	PM, Research Lead, etc.
notes	string	No	
PricingPack			
Field	Type	Req.	Notes
pricing_pack_id	string (UUID)	Yes	Primary key.
opp_id	string (UUID)	Yes	FK to Opportunity.
label	enum	Yes	BASE | GOOD | BETTER | BEST | CUSTOM.
currency	string	Yes	ISO-4217, e.g., USD.
is_selected	bool	Yes	Marks the pack used for submission.
version	int	Yes	Increment on changes.
created_at	datetime	Yes	

updated_at	datetime	Yes	
FieldEvidence			
Field	Type	Req.	Notes
evidence_id	string (UUID)	Yes	Primary key.
opp_id	string (UUID)	Yes	
field_path	string	Yes	e.g., brief.loi_minutes
source_type	enum	Yes	email_message | attachment | manual_entry.
source_id	string (UUID)	No	EmailMessage.message_id or Attachment.attachment_id when applicable.
snippet_text	string	No	Small excerpt used in UI.
page_number	int	No	Attachment page when applicable.
char_start	int	No	Optional offsets for highlighting.
char_end	int	No	
confidence	enum	No	low | medium | high.
created_at	datetime	Yes	
created_by	enum	Yes	system_parser | user.

Stores provenance linking parsed fields to email/attachment sources (supports audit and user trust).

FieldEvidence

Optional container to support Good/Better/Best or alternate commercial offers. PricingLine and PricingSummary can reference pricing_pack_id.

PricingPack

MilestoneTemplate (used inside WorkplanTemplate)

Default milestone template used to seed Workplan for a given study type.

WorkplanTemplate

Groups ClarificationQuestionDefinition IDs for a study type (global default or tenant override).

QuestionSet

Groups MultiplierDefinition codes for a study type (global default or tenant override).

MultiplierSet

These entities close v1.0 reference gaps and make study scoping reproducible: sets group library items; templates drive default workplans; provenance links parsed fields to sources.

8. Rules and estimation engine

VERITY SCOPE™ uses a deterministic rules engine. The product can use an LLM to draft text (e.g., proposal narrative), but all calculation and gating logic is rule-based and auditable.

8.1 Rule evaluation context

- opportunity: metadata (study_type, due_date, owner, status).
- brief: structured inputs (countries, audiences, sample, LOI, deliverables).
- scope: selected design, sample plan, deliverables, assumptions.
- tenant_config: rate cards, pricing policies, templates, vendors.
- pricing: calculated totals and margin checks (after costing).
- brief.countries_count = len(brief.countries)
- brief.languages_count = len(brief.languages) (defaults to 1 if not provided)
- brief.audience_count = len(brief.audiences)
- brief.is_multi_country = (brief.countries_count > 1)
- opportunity.days_to_due = days between today and opportunity.due_date

Derived fields (computed by platform for rule evaluation)

8.2 Expression language

For production, use a safe expression language (no arbitrary code execution). Examples: JSONLogic, CEL, or a restricted custom DSL. Expressions must be versioned so historical bids remain reproducible.

8.3 Task selection and effort calculation

- Task selection: include TaskDefinition if applies_when evaluates to true.
- Units calculation: units_expr returns a number (e.g., countries_count * languages_count).
- Base hours: per unit, per role. Each task can produce multiple WBS lines (one per role).
- Multipliers: multiply hours by all applicable multipliers for that task code prefix.
- Output: a WBSLine list with full traceability to the task library and multiplier log.

8.4 JSON templates (canonical)

These templates are intended for developers to copy-paste as a starting point.

```
TaskDefinition (example)

{
  "task_code": "QNT-PROG-PROJECT_SETUP",
  "name": "Create survey project shell",
  "phase": "programming",
  "applies_when": "scope.method == 'quant'",
  "unit_type": "per_country_language",
  "units_expr": "brief.countries_count * brief.languages_count",
  "base_hours_by_role": { "Programmer": 1.5 },
  "multiplier_codes": ["M_LOI", "M_RUSH"],
  "dependencies": ["QNT-INS-DRAFT_CORE"],
  "outputs": ["survey_shell_v1"],
  "active": true
}

MultiplierDefinition (example)

{
  "multiplier_code": "M_LOI",
  "driver": "brief.loi_minutes",
  "levels": [
```

```
    { "when": "brief.loi_minutes <= 10", "value": 1.0 },
    { "when": "brief.loi_minutes <= 20", "value": 1.15 },
    { "when": "brief.loi_minutes <= 30", "value": 1.3 },
    { "when": "brief.loi_minutes > 30", "value": 1.5 }
  ],
  "apply_to_task_prefixes": ["QNT-INS", "QNT-PROG", "QNT-DAT"],
  "active": true
}
```

ClarificationQuestionDefinition (example)

```
{
  "question_id": "UAA-QUANT-LOI-01",
  "study_type": "UAA",
  "method": "quant",
  "section": "Instrument",
  "critical": true,
  "question_text": "What is the target LOI (minutes)?",
  "mapped_field": "brief.loi_minutes",
  "trigger_when": "brief.loi_minutes == null",
  "default_assumption": "15",
  "active": true
}
```

9. PMR studies library (MECE v2)

This library is designed to be MECE at the family level (each project has one primary family). Study types sit under families. Orthogonal tags (audience, mode, analytics modules) are applied separately.

9.1 Study families (primary classification)

- Understanding & Diagnosis (descriptive): U&A / needs / barriers / journeys.
- Tracking & Monitoring (longitudinal): trackers and wave-based monitoring.
- Testing & Optimization: concept, positioning, messaging, materials testing.
- Trade-off & Choice Modeling: conjoint/DCE, maxdiff, related choice models.
- Segmentation & Targeting: build, validate, size, and activate segments.
- Pricing & Market Access: WTP studies plus payer/HTA decision research.

9.2 Orthogonal tags (not separate families)

- Audience tag: HCP, patient, caregiver, payer/HTA, pharmacist, nurse, admin, KOL/expert.
- Mode tag: CAWI, CATI, CAPI, IDI, FGD, pair/triad, asynchronous community, diary, digital ethnography/co-creation.
- Analytics module tag: key drivers, segmentation, conjoint/maxdiff, TURF/portfolio, market structure, simulator outputs.
- Recruit difficulty tag: easy | medium | hard | rare.

9.3 Study types (default set)

Study family	Study type	Default method	Common modes	Key drivers	Default deliverables (typical)
Understanding & Diagnosis	U&A / ATU (Ad-hoc)	Quant	CAWI/CATI	countries, sample_n, LOI, quotas	Tabbook + topline + report + readout
Understanding & Diagnosis	Needs & Barriers	Qual or Mixed	IDI/FGD/Community	sessions, markets, stimuli, recruit difficulty	Insight report + quotes + workshop
Understanding & Diagnosis	Journey Mapping	Qual or Mixed	IDI/Diary	sessions, journey depth, transcripts	Journey map + personas + workshop
Tracking & Monitoring	Brand/HCP/Patient Tracker	Quant	CAWI/CATI	waves, sample_n, LOI, trend outputs	Trend deck + dashboard option
Tracking & Monitoring	Message/Ad Tracking (as tracker subtype)	Quant	CAWI	waves, sample_n, stimuli	Trend deck + KPIs
Testing & Optimization	Concept Test	Quant	CAWI	concepts, sample_n, LOI, allocation	Scorecard + recommendations
Testing & Optimization	Positioning / Value Proposition Test	Quant or Mixed	CAWI/IDI	claims, audiences, geos	Positioning recommendation deck
Testing &	Message / Claim	Quant	CAWI/IDI	messages,	Message

Optimization	Test	or Mixed		stimuli, compliance	hierarchy + revisions
Testing & Optimization	Materials / Creative Test	Quant or Mixed	CAWI/IDI	stimuli formats, device, sample	Creative diagnostics + next steps
Trade-off & Choice	MaxDiff	Quant	CAWI	items, sample_n, segments	Priority list + subgroup cuts
Trade-off & Choice	Conjoint / DCE	Quant	CAWI	attributes/levels, sample_n, simulator	Utilities + simulator scenarios
Segmentation & Targeting	Segmentation Build	Quant	CAWI	sample_n, variables, segments	Segment personas + sizing + assignment
Segmentation & Targeting	Segmentation Validation/Sizing	Quant	CAWI	sample_n, classification model	Sizing + confidence + rules
Pricing & Market Access	Pricing / WTP (VWD/GG)	Quant	CAWI/CATI	price points, sample_n, audience	Price corridors + elasticity outputs
Pricing & Market Access	Payer / HTA Decision Research	Qual or Mixed	IDI	markets, roles, compliance	Access archetypes + evidence needs
Understanding & Diagnosis	KOL / Expert Interviews (tagged IDI)	Qual	IDI	sessions, recruit difficulty, compliance	Expert themes + implications
Special handling (tag)	Rare disease / Hard-to-reach	Any	Any	incidence, recruit sources, timeline	Feasibility + assumptions emphasized

10. Email-first intake and submission

10.1 Inbound email intake

- Connect mailbox (preferred) or use a forwarding alias (low-IT option).
- On new inbound email: create/attach to Opportunity, store thread, store attachments.
- Parse attachments and email body into structured Brief fields with confidence levels.
- Link parsed fields back to source snippets for auditability (FieldEvidence records; internal).
- Run completeness scoring and generate gaps list.

10.2 Clarifications and iterations

- Clarifications are created as ClarificationInstance records and grouped into a client-facing email draft.
- Each clarification is mapped to a specific field (e.g., brief.loi_minutes).
- If client replies, the agent can re-parse the reply and auto-resolve questions when confidence is high.
- If client does not reply before the internal cut-off, the system applies default assumptions (must be explicitly stated).

10.3 Outbound submission (default mode)

- Generate a submission-ready email draft with the correct attachments and naming conventions.
- Human user reviews and sends from their email client (lowest friction).
- System logs the final submitted package and timestamp once the user confirms sent status.
- Optional enterprise mode: send via connected mailbox after approvals.

10.4 Email safety and governance requirements

- Tenant-configurable allowed domains (to prevent accidental external sends).
- Attachment size limits and malware scanning (if supported by deployment).
- Redaction controls for sensitive internal pricing notes (internal-only fields not exported).
- Full audit trail: who generated draft, who approved, who sent.

11. Integrations (CRM, procurement, storage)

11.1 CRM integration (Salesforce / Dynamics / HubSpot)

CRM integration is optional and most useful for vendors/agencies managing a pipeline of opportunities. VERITY SCOPE™ should treat the CRM as a system of record for pipeline metadata, while VERITY SCOPE™ remains the system of record for bid artifacts.

Minimum CRM field mapping (recommended)

CRM field	Verity SCOPE field	Direction
Opportunity ID	opportunity.source_ref	CRM -> SCOPE
Account/Client	client_account.client_name	CRM -> SCOPE
Opportunity name	opportunity.title	CRM -> SCOPE
Owner	opportunity.owner_user_id	CRM -> SCOPE
Due date / close date	opportunity.due_date	CRM -> SCOPE
Stage	opportunity.status (mapped)	SCOPE -> CRM
Amount (optional)	pricing_summary.total_price (optional)	SCOPE -> CRM
Notes/activity	audit events (summary)	SCOPE -> CRM

11.2 Procurement integration (e.g., Coupa)

Procurement integration is optional and most useful for pharma/internal PMR teams where vendor engagements require requisitions, approvals, and purchase orders. VERITY SCOPE™ should treat the procurement system as the system of record for PO and spend controls, while keeping scope and WBS traceability in VERITY SCOPE™.

Minimum procurement mapping (recommended)

Procurement field	Verity SCOPE field	Direction
Supplier name	vendor.name	SCOPE -> Procurement
Contract/SoW attachment	document_artifact(file_uri)	SCOPE -> Procurement
PO number	handoff_pack.po_number (optional extension)	Procurement -> SCOPE
Approved budget	pricing_summary.total_price	SCOPE -> Procurement
Cost center/project code	opportunity.metadata.cost_center	Procurement -> SCOPE / bid intake

11.3 Storage and document systems

- Object storage (required): store raw intake files and generated artifacts (S3/GCS/Azure Blob equivalent).
- Optional: SharePoint/Google Drive export for client-facing folders.
- Naming conventions should include client, opp_id, doc_type, and version.

11.4 Identity and access (SSO)

- Support SSO (SAML/OIDC) for enterprise tenants.
- Support SCIM provisioning where required.
- Enforce least-privilege RBAC for approvals and submissions.

12. Security, compliance, and audit

12.1 Data classification (recommended)

- Public: product documentation and non-client-specific sample templates.
- Confidential: bid content, pricing, internal notes, vendor quotes.
- Restricted: personally identifiable information (PII) and any patient-related details (avoid storing unless necessary).

12.2 Security controls (baseline)

- Encryption in transit (TLS) and at rest (KMS-managed).
- Tenant isolation at the data layer (row-level + storage prefixes).
- RBAC for data access and action permissions (approve, submit, configure).
- Immutable audit trail for key actions (scope changes, pricing changes, approvals, submission).
- Configurable retention and deletion policies per tenant.

12.3 PMR-specific compliance considerations

- If patient or HCP contact data is included in briefs, enforce redaction or restricted storage policies.
- Support documentation of incentives and fair market value constraints in the assumptions register.
- Support an explicit adverse event handling process tag where relevant (for certain patient-facing work).
- Ensure client-facing exports do not include internal-only notes or margin details unless explicitly enabled.

12.4 Audit events to log (minimum)

- Opportunity created, brief parsed, brief edited.
- Clarifications generated/sent, assumptions applied.
- Scope version created/updated, WBS generated, pricing generated.
- Approvals requested/approved/rejected.
- Documents generated, QC pass/fail.
- Submission prepared/sent, win/loss recorded, handoff created.

13. Non-functional requirements

13.1 Performance and scalability

- Parse time: typical email + attachments should be processed in minutes, not hours.
- Estimation recalculation: WBS + pricing recompute should complete in seconds for interactive edits.
- Support concurrent bids and multi-user collaboration per tenant.

13.2 Reliability and error handling

- Every step is resumable (idempotent processing on retries).
- Clear user-facing error messages for parsing failures and missing configuration (e.g., rate card not configured).
- No silent failures for approvals or submission steps.

13.3 Observability

- Central logs with correlation IDs per opportunity.
- Metrics: cycle time by status, rework cycles, approval SLA adherence, parsing success rate.
- Alerts for integration failures (email connector, CRM sync, storage).

13.4 Reproducibility

- Version all libraries (study definitions, task sets, multipliers, question sets).
- Store which version was used for each opportunity so the same bid can be reproduced later.
- Do not change historical bids when global libraries evolve.

14. MVP scope and roadmap

14.1 MVP (practical, email-first)

- Email intake: connect mailbox or forwarding alias; store thread + attachments.
- Brief parser: extract key fields + completeness score; allow manual edits.
- Study library v1: MECE families and core study types; playbook selection.
- Clarifications engine: question bank + assumptions register + email draft.
- Scope builder: sample plan grid, deliverables selection, timeline template.
- Estimation engine: task library + units + multipliers + WBS generation.
- Tenant configuration: rate card upload, basic pricing policy thresholds.
- Pricing pack: labor + OOP baseline + margin check + exceptions.
- Doc generation: proposal and pricing annex (template-based).
- Approval gates: scope + pricing approvals; audit log.
- Submission: email draft with attachments; submission record.

14.2 Roadmap (high-value additions)

- CRM sync: create/update opportunities and attach artifacts.
- Procurement sync: requisitions/PO metadata and SoW attachments.
- Template editor UI and clause library management.
- Historical benchmark learning: recommend base hours from past projects (still deterministic at runtime).
- Delivery handoff integrations: create project plans in delivery tools and track scope changes.

Appendix A. Enumerations

A1. Opportunity status

- NEW_INTAKE
- BRIEF_PARSED
- BRIEF_GAPS_FOUND
- CLARIFICATIONS_SENT
- SCOPE_DRAFTED
- WBS_GENERATED
- COSTED
- DOCS_DRAFTED
- INTERNAL_REVIEW

REWORK_REQUIRED

- APPROVED_TO_SUBMIT
- SUBMITTED
- WON_HANDOFF
- LOST_ARCHIVE
- CANCELLED

A2. Study families (primary)

- UNDERSTANDING_DIAGNOSIS
- TRACKING_MONITORING
- TESTING_OPTIMIZATION
- TRADEOFF_CHOICE
- SEGMENTATION_TARGETING
- PRICING_MARKET_ACCESS

A3. Study types (default v1)

- UAA_ADHOC
- NEEDS_BARRIERS
- JOURNEY_MAPPING
- TRACKER
- MESSAGE_AD_TRACKING
- CONCEPT_TEST
- POSITIONING_TEST
- MESSAGE_CLAIM_TEST
- MATERIALS_CREATIVE_TEST
- MAXDIFF
- CONJOINT_DCE
- SEGMENTATION_BUILD
- SEGMENTATION_VALIDATION

- PRICING_WTP
- PAYER_HTA
- KOL_EXPERT_IDI

A4. Methods

- quant
- qual
- mixed

A5. Data collection modes (tags)

- CAWI
- CATI
- CAPI
- IDI
- FGD
- PAIR
- TRIAD
- ASYNC_COMMUNITY
- DIARY
- DIGITAL_ETHNOGRAPHY
- CO_CREATION

A6. Standard deliverables (sample set)

- TABBOOK
- TOPLINE_DECK
- FULL_REPORT
- EXEC_SUMMARY
- READOUT_SESSION
- WORKSHOP
- DASHBOARD
- RAW_DATA_EXPORT
- CODEBOOK
- TRANSCRIPTS
- QUOTE_BANK

A7. HCP shortlist status

- draft
- approved

Appendix B. JSON templates (developer copy-paste)

These are not exhaustive JSON Schemas; they are canonical payload examples for implementation.

```
Opportunity (example)

{
  "opp_id": "uuid",
  "tenant_id": "uuid",
  "client_id": "uuid",
  "title": "HCP U&A - Oncology - 3 markets",
  "owner_user_id": "uuid",
  "status": "BRIEF_PARSED",
  "due_date": "2026-03-10",
  "study_family": "UNDERSTANDING_DIAGNOSIS",
  "study_type": "UAA_ADHOC",
  "method": "quant",
  "priority": "P1",
  "source": "email",
  "source_ref": "email_message_id",
  "version": 1
}

Brief (example)

{
  "brief_id": "uuid",
  "opp_id": "uuid",
  "objectives": ["Understand current prescribing drivers", "Identify unmet needs"],
  "audiences": [
    {
      "audience_type": "hcp",
      "name": "Oncologists",
      "inclusion_criteria": ["Treat >= 10 patients/month"],
      "recruit_difficulty": "medium"
    }
  ],
  "countries": ["US", "DE", "JP"],
  "languages": ["en", "de", "ja"],
  "method_preference": "quant",
  "sample_total_target": 450,
  "sample_by_country": { "US": 200, "DE": 150, "JP": 100 },
  "loi_minutes": 18,
  "advanced_analytics": ["drivers"],
  "deliverables_requested": ["TABBOOK", "FULL_REPORT", "READOUT_SESSION"],
  "fieldwork_window_start": "2026-03-15",
  "fieldwork_window_end": "2026-03-28",
  "completeness_score": 82,
  "confidence_by_field": { "loi_minutes": "high", "sample_total_target": "medium" }
}

SampleSizeRecommendation (example)
```

```
{
  "recommendation_id": "uuid",
  "opp_id": "uuid",
  "options": [
    {
      "option_code": "minimum",
      "total_n": 60,
      "timeline_impact_days": 0,
      "cost_impact_percent": -15,
      "assumptions_used": ["uuid"]
    },
    {
      "option_code": "recommended",
      "total_n": 90,
      "timeline_impact_days": 7,
      "cost_impact_percent": 0,
      "assumptions_used": ["uuid"]
    },
    {
      "option_code": "high_confidence",
      "total_n": 140,
      "timeline_impact_days": 14,
      "cost_impact_percent": 20,
      "assumptions_used": ["uuid"]
    }
  ],
  "selected_option": "recommended",
  "confidence_score": 0.78,
  "rationale": "Based on segments x countries and feasibility in the HCP universe.",
  "evidence": [{ "evidence_id": "uuid", "field_path": "brief.sample_total_target" }],
  "created_at": "2026-02-26T00:00:00Z",
  "updated_at": "2026-02-26T00:00:00Z"
}
```


HCPUniverseQuery (example)


```
{
  "query_id": "uuid",
  "opp_id": "uuid",
  "audience_name": "Oncologists",
  "countries": ["US"],
  "filters": {
    "specialty": ["Oncology"],
    "setting": ["Hospital", "Clinic"],
    "min_patient_volume_per_month": 20
  },
  "exclusions": {
    "blacklist_ids": [],
    "compliance_blocks": []
  },
  "result_counts": {
    "US": { "eligible_total": 860, "by_setting": { "Hospital": 520, "Clinic": 340 } }
  },
  "evidence": [{ "evidence_id": "uuid", "field_path": "brief.audiences[0]" }],
  "created_at": "2026-02-26T00:00:00Z",
  "updated_at": "2026-02-26T00:00:00Z"
}
```

HCPShortlist (example)


```
{
  "shortlist_id": "uuid",
  "opp_id": "uuid",
  "audience_name": "Oncologists",
  "countries": ["US"],
  "target_n": 90,
  "primary_hcp_ids": ["hcp_001", "hcp_002", "hcp_003"],
  "backup_hcp_ids": ["hcp_101", "hcp_102"],
  "scoring_model_version": "v1.0",
  "selection_rationale": "Ranked by fit and past reliability; quota-balanced across
settings.",
  "status": "draft",
  "approved_by_user_id": null,
  "approved_at": null,
  "created_at": "2026-02-26T00:00:00Z",
  "updated_at": "2026-02-26T00:00:00Z"
}
```


WBSLine (example)


```
{
  "wbs_line_id": "uuid",
  "opp_id": "uuid",
  "task_code": "QNT-PROG-PROJECT_SETUP",
  "role": "Programmer",
  "unit_type": "per_country_language",
  "units": 3,
  "base_hours": 1.5,
  "multipliers_applied": [{ "multiplier_code": "M_RUSH", "value": 1.2 }],
  "total_hours": 5.4
}
```


PricingSummary (example)


```
{
  "opp_id": "uuid",
  "currency": "USD",
  "labor_price_total": 52000,
  "oop_price_total": 18000,
  "total_price": 70000,
  "gross_margin_percent": 38.5,
  "discount_percent": 0,
  "exceptions": []
}
```

Appendix C. Example bid package payload

This payload represents the logical 'Bid Package' that can be exported, stored, or attached to CRM/procurement systems. It bundles the final versions of scope, workplan, WBS, pricing, and documents.

```
BidPackage (example)

{
  "opp_id": "uuid",
  "scope_version": 2,
  "workplan_version": 1,
  "pricing_version": 2,
  "scope": { "...": "..." },
  "workplan": { "...": "..." },
  "wbs_lines": [{ "...": "..." }],
  "pricing_summary": { "...": "..." },
  "documents": [
    { "doc_type": "proposal", "version": 2, "file_uri": "s3://.../proposal_v2.docx" },
    { "doc_type": "pricing_annex", "version": 2, "file_uri":
"s3://.../pricing_annex_v2.xlsx" }
  ],
  "assumptions_risks": [{ "...": "..." }],
  "approvals": [{ "...": "..." }],
  "submission": { "...": "..." }
}
```